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Display panel, method for manufacturing the same and display device

Abstract

Provided is a display panel, including: a substrate provided with a display region and a non-display region disposed at a periphery of the display region; a plurality of first power signal lines and a plurality of drive signal lines that are disposed in the display region; and a power connection bus, a plurality of first fan-out leads, and a plurality of second fan-out leads that are disposed in the non-display region, wherein the power connection bus is electrically connected to the first power signal line and the first fan-out lead respectively; the plurality of second fan-out leads are electrically connected to the plurality of drive signal lines in a one-to-one correspondence; each first fan-out lead is disposed between every two adjacent second fan-out leads; and the plurality of first fan-out leads and the plurality of second fan-out leads are both configured to be electrically connected to a driving chip.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a 371 of PCT Application PCT/CN2021/077045 filed Feb. 20, 2021, the disclosure of which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the field of display technologies, and in particular relates to a display panel, a method for manufacturing the same, and a display device.

BACKGROUND

(3) With the development of display technologies, display devices are used more and more widely. Display devices usually include smartphones, tablet computers, televisions, monitors, and the like.

(4) Generally, a display device includes a display panel and a driving chip. The display panel is provided with a display region and a non-display region. The display panel includes a plurality of sub-pixels and signal lines electrically connected to the plurality of sub-pixels that are disposed in the display region, and a lead disposed in the non-display region. The lead is electrically connected to a plurality of signal lines and the driving chip, respectively. In this way, the driving chip controls the plurality of sub-pixels via the lead and the plurality of signal lines, such that the display panel can display a picture.

(5) However, a non-display region in the currently used display panel is larger in width, which results in a lower screen-to-body ratio of the display device, which affects the display effect of the display device.

SUMMARY

(6) Embodiments of the present disclosure provide a display panel, a method for manufacturing the same, and a display device. In the prior art, a display device has a low screen-to-body ratio, which affects the display effect of the display device. The problem can be solved by the following technical solutions:

(7) According to a first aspect, a display panel is provided. The display panel includes:

(8) a substrate provided with a display region and a non-display region disposed at a periphery of the display region;

(9) a plurality of first power signal lines and a plurality of drive signal lines that are disposed in the display region; and

(10) a power connection bus, a plurality of first fan-out leads, and a plurality of second fan-out leads that are disposed in the non-display region, wherein

(11) the power connection bus is electrically connected to the first power signal line and the first fan-out lead respectively; the plurality of second fan-out leads are electrically connected to the plurality of drive signal lines in a one-to-one correspondence; each first fan-out lead is disposed between every two adjacent second fan-out leads; and the plurality of first fan-out leads and the plurality of second fan-out leads are both configured to be electrically connected to a driving chip.

(12) Optionally, one first fan-out lead corresponds to n first power signal lines; and the first fan-out lead is configured to apply a first power signal to the n first power signal lines corresponding to the first fan-out lead via the power connection bus, wherein n is an integer greater than or equal to 1.

(13) Optionally, on the power connection bus, a number of second connection points distributed on a side of a first connection point is equal to a number of second connection points distributed on another side of the first connection point, wherein the first connection point is a connection point for connecting the first fan-out lead to the power connection bus; and the second connection point is a connection point for connecting the first power signal line corresponding to the first fan-out lead to the power connection bus.

(14) Optionally, n is equal to 4.

(15) Optionally, a ratio of a width of the power connection bus to a maximum width of the first fan-out lead ranges from 0.7 to 1.3.

(16) Optionally, the first fan-out lead includes a first sub-lead and a second sub-lead electrically

connected to the first sub-lead; wherein the first sub-lead and the first power signal line are disposed at a same layer, and the second sub-lead and the power connection bus are disposed at a same layer, the first power signal lines and the power connection bus being disposed at different layers.

(17) Optionally, an orthographic projection of the first sub-lead on the substrate falls within an orthographic projection of the second sub-lead on the substrate.

(18) Optionally, two adjacent second fan-out leads are disposed at different layers; and orthographic projections, on the substrate, of the two adjacent second fan-out leads disposed at different layers are staggered.

(19) Optionally, the plurality of second fan-out leads includes a plurality of third sub-leads and a plurality of fourth sub-leads, wherein the plurality of third sub-leads and the plurality of drive signal lines are disposed at a same layer, the plurality of fourth sub-leads and the plurality of drive signal lines are disposed at different layers, the plurality of third sub-leads and the plurality of fourth sub-leads being arranged in a staggered fashion.

(20) Optionally, the display panel further includes an antistatic structure disposed in the non-display region, the antistatic structure being disposed on one side, distal from the power connection bus, of the display region.

(21) Optionally, the display panel further includes a plurality of columns of pixels disposed in the display region, each column of pixels include at least two columns of sub-pixels, the plurality of columns of pixels are in a one-to-one correspondence with the plurality of first power signal lines, and the first power signal line is electrically connected to a plurality of sub-pixels in a corresponding column of pixels; and the plurality of drive signal lines include: a plurality of data signal lines in a one-to-one correspondence with a plurality of columns of sub-pixels in the display panel and a plurality of sense signal lines in a one-to-one correspondence with the plurality of columns of pixels, wherein the data signal line is electrically connected to a plurality of sub-pixels in a corresponding column of sub-pixels; and the sense signal line is electrically connected to a plurality of sub-pixels in a corresponding column of pixels.

(22) Optionally, the display panel further includes a first connection line and a second connection line that are disposed in the display region; wherein the first connection line and the first power signal line are disposed at different layers; and the second connection line and the sense signal line are disposed at different layers; the first power signal line being electrically connected to the plurality of sub-pixels in the corresponding column of pixels via the first connection line, and the sense signal line being electrically connected to the plurality of sub-pixels in the corresponding column of pixels via the second connection line.

(23) Optionally, the display panel further includes a second power signal line disposed in the non-display region, wherein the second power signal line is disposed on one side, proximal to the power connection bus, of the display region, the second power signal line being configured to be electrically connected to the driving chip.

(24) Optionally, the plurality of data signal lines include a plurality of groups of data signal lines, wherein each group of data signal lines include two adjacent data signal lines, and one first power signal line or one sense signal line is disposed between any two adjacent groups of data signal lines.

(25) Optionally, the drive signal line and the power connection bus are disposed at different layers, wherein the drive signal line includes a drive signal line body and a jumper signal line connected to the drive signal line body, an overlapping area exists between an orthographic projection of the jumper signal line on the substrate and an orthographic projection of the power connection bus on the substrate, and a width of the jumper signal line is larger than a width of the drive signal line body.

(26) Optionally, an extension direction of the first power signal line is parallel to an extension direction of the drive signal line, and is perpendicular to an extension direction of the power

connection bus.

(27) According to a second aspect, a method for manufacturing a display panel is provided. The method includes: providing a substrate, wherein the substrate is provided with a display region and a non-display region disposed at the periphery of the display region; and forming a plurality of first power signal lines and a plurality of drive signal lines in the display region, and forming a power connection bus, a plurality of first fan-out leads and a plurality of second fan-out leads in the non-display region, wherein the power connection bus is electrically connected to the first power signal line and the first fan-out lead respectively; the plurality of second fan-out leads are electrically connected to the plurality of drive signal lines in a one-to-one correspondence; each first fan-out lead is disposed between every two adjacent second fan-out leads; and the plurality of first fan-out leads and the plurality of second fan-out leads are both configured to be electrically connected to a driving chip.

(28) According to a third aspect, a display device is provided. The display device includes: a driving chip and the display panel as described above, wherein the driving chip is electrically connected to the plurality of first fan-out leads and the plurality of second fan-out leads in the display panel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) To describe the technical solutions in the embodiments of the present disclosure more clearly, the following briefly introduces the accompanying drawings required for describing the embodiments. Apparently, the accompanying drawings in the following description show merely some embodiments of the present disclosure, and a person of ordinary skill in the art may derive other drawings from these accompanying drawings without creative efforts.

(2) FIG. 1 is a schematic structural diagram of a display device according to the related art;

(3) FIG. 2 is a schematic structural diagram of a display panel according to an embodiment of the present disclosure;

(4) FIG. 3 is a partially enlarged view of the display panel shown in FIG. 2 at A;

(5) FIG. 4 is a schematic structural diagram of a display panel according to another embodiment of the present disclosure;

(6) FIG. 5 is a partially enlarged view of the display panel shown in FIG. 4 at B;

(7) FIG. 6 is a schematic structural diagram of films of the display panel shown in FIG. 4;

(8) FIG. 7 is a partially enlarged view of the display panel shown in FIG. 4 at C;

(9) FIG. 8 is a schematic diagram of a pixel drive circuit according to an embodiment of the present disclosure;

(10) FIG. 9 is a schematic structural diagram of an antistatic structure according to an embodiment of the present disclosure;

(11) FIG. 10 is a partially enlarged view of the display panel shown in FIG. 5 at D;

(12) FIG. 11 is a schematic structural diagram of a display device according to an embodiment of the present disclosure;

(13) FIG. 12 is an effect diagram of a display device according to the related art;

(14) FIG. 13 is an effect diagram of a display device according to an embodiment of the present disclosure;

(15) FIG. 14 is an effect diagram of a splicing screen according to the related art; and

(16) FIG. 15 is an effect diagram of a splicing screen according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(17) For clearer descriptions of the objectives, technical solutions, and advantages of the present

disclosure, embodiments of the present disclosure are described in detail hereinafter with reference to the accompanying drawings.

(18) In the related art, referring to FIG. 1, which is a schematic structural diagram of a display device according to the related art. The display device includes a display panel **00** and a driving chip **10**. The display panel **00** is provided with a display region **00a** and a non-display region **00b**. The display panel **00** includes: a plurality of sub-pixels (not shown), a plurality of power signal lines **01** (thick solid lines in the figure), and a plurality of drive signal lines (not shown) that are disposed in the display region **00a**, wherein the plurality of power signal lines **01** and the plurality of drive signal lines are connected to the plurality of sub-pixels, and; and a power connection bus **02**, two first leads **03**, and a plurality of second leads (not shown) that are disposed in the non-display region **00b**.

(19) The plurality of second leads are electrically connected to the plurality of drive signal lines in a one-to-one correspondence, a region where the plurality of second leads are disposed may also be referred to as a fan-out region **00c**, and each of the plurality of second leads is electrically connected to the driving chip **10**. The power connection bus **02** is electrically connected to the power signal line **01** and the first lead **03** respectively, the two first leads **03** are respectively disposed on two sides of the fan-out region **00c**, and the two first leads **03** are electrically connected to the driving chip **10**. In this way, the driving chip **10** can apply a power signal to the plurality of power signal lines **01** via the two first leads **03** and the power connection bus **02** and apply a drive signal to the plurality of drive signal lines via the plurality of second leads, so as to control the sub-pixels that are connected to the plurality of power signal lines **01** and the plurality of drive signal lines for display by the display panel.

(20) In the related art, the power connection bus **02** is electrically connected to the driving chip **10** only via the two first leads **03**, and therefore the power connection bus **02** needs to carry a large current. In this way, a width of the power connection bus **02** is large, which results in a larger width of the non-display region **00b** in the display panel **00**, and further results in a larger width of the frame of the display panel **00**. Thus, a screen-to-body ratio of a display device prepared by using the display panel **00** is low, thus affecting the display effect of the display device.

(21) Referring to FIG. 2, FIG. 2 is a schematic structural diagram of a display panel according to an embodiment of the present disclosure. The display panel **000** may be an organic light-emitting display (OLED) panel, an active matrix organic light-emitting diode (AM-OLED) display panel or a liquid crystal display panel. In the case where the display panel **000** is an OLED display panel or an AM-OLED display panel, the display panel **000** may be a top-emission display panel or a bottom-emission display panel.

(22) The display panel **000** includes a substrate **100**, wherein the substrate **100** is provided with a display region **100a** and a non-display region **100b** disposed at a periphery of the display region **100a**. To show the structure of the non-display region **100b** more clearly, referring to FIG. 3, which is a partially enlarged view of the display panel shown in FIG. 2 at A.

(23) The display panel **000** further includes a plurality of first power signal lines **200** (thick solid lines in the figure) and a plurality of drive signal lines **300** (thick dashed lines in the figure) that are disposed in the display region **100a**, and a power connection bus **400**, a plurality of first fan-out leads **500** (thin solid lines in the figure) and a plurality of second fan-out leads **600** (thin dashed lines in the figure) that are disposed in the non-display region **100b**.

(24) The power connection bus **400** is electrically connected to the first power signal line **200** and the first fan-out lead **500** respectively, the plurality of second fan-out leads **600** are electrically connected to the plurality of drive signal lines **300** in a one-to-one correspondence, and any one of the plurality of first fan-out leads **500** is disposed between two adjacent second fan-out leads **600**. The plurality of first fan-out leads **500** and the plurality of second fan-out leads **600** are both configured to be electrically connected to a driving chip.

(25) It should be noted that there are usually several hundred or thousand fan-out leads **500** and

second fan-out leads **600** in the display panel **000**, and there may be a plurality of second fan-out leads **600** between two adjacent first fan-out leads **500**.

(26) In the present disclosure, any one of the first fan-out leads **500** is disposed between two adjacent second fan-out leads **600**, and a region in which the plurality of second fan-out leads **600** are disposed may also be referred to as a fan-out region **100c**. Therefore, the plurality of first fan-out leads **500** are also disposed in the fan-out region **100c**. The power connection bus **400** in the display panel **000** is electrically connected to the driving chip via the plurality of first fan-out leads **500** disposed in the fan-out region **100c**. In this way, the current carried by the power connection bus **400** is reduced, such that a width of the power connection bus **400** is reduced, a width of the non-display region **000b** in the display panel **000** is reduced, and further a width of the frame of the display panel **000** is reduced, thereby increasing a screen-to-body ratio of the display device prepared by using the display panel **000** and improving the display effect of the display device.

(27) In summary, the display panel according to the embodiment of the present disclosure includes a substrate, a plurality of first power signal lines, a plurality of drive signal lines, a power connection bus, a plurality of first fan-out leads, and a plurality of second fan-out leads. Any one of the first fan-out leads in the display panel is disposed between two adjacent second fan-out leads, and a region in which the plurality of second fan-out leads are disposed may also be referred to as a fan-out region. Therefore, the first fan-out leads are also disposed in the fan-out region, and the power connection bus in the display panel is electrically connected to the driving chip via the plurality of first fan-out leads disposed in the fan-out region. In this way, the current carried by the power connection bus is reduced, such that a width of the power connection bus is reduced, a width of the non-display region in the display panel is reduced, and further a width of the frame of the display panel is reduced, thereby increasing a screen-to-body ratio of the display device prepared by using the display panel and improving the display effect of the display device.

(28) In an embodiment of the present disclosure, referring to FIG. 4 and FIG. 5, FIG. 4 is a schematic structural diagram of a display panel according to another embodiment of the present disclosure, and FIG. 5 is a partially enlarged view of the display panel shown in FIG. 4 at B. In the display panel **000**, one first fan-out lead **500** corresponds to n first power signal lines **200**, wherein n is an integer greater than or equal to 1. The first fan-out lead **500** is configured to apply a first power signal to the n first power signal lines **200** corresponding to the first fan-out lead **500** via the power connection bus **400**. In this way, a number of the first fan-out leads **500** disposed in the non-display region **100b** is less than a number of the first power signal lines **200** disposed in the display region **100a**, that is to say, the number of the first fan-out leads **500** disposed in the non-display region **100b** is effectively reduced, thereby further reducing the width of the non-display region **100b** in the display panel **000** and increasing the screen-to-body ratio of the display device prepared by using the display panel **000**.

(29) Optionally, a connection point for connecting each first fan-out lead **500** to the power connection bus **400** is a first connection point **400a**, and a connection point for connecting the first power signal line **200** corresponding to the first fan-out lead **500** to the power connection bus **400** is a second connection point **400b**. On the power connection bus **400**, a number of the second connection points **400b** distributed on a side of each first connection point **400a** is equal to a number of the second connection points **400b** distributed on another side of the first connection point **400a**. In this way, n first power signal lines **200** corresponding to each first fan-out lead **500** are uniformly distributed on two sides of the first fan-out lead **500**, such that the first power signal applied by the first fan-out lead **500** to the n first power signal lines **200** corresponding to the first fan-out lead **500** via the power connection bus **400** has a better uniformity.

(30) For example, n may be equal to 4, that is, one first fan-out lead **500** corresponds to four first power signal lines **200**.

(31) In the related art, as shown in FIG. 1, as the power connection bus **02** is electrically connected to the driving chip **10** only via two first leads **03**, the power connection bus **02** needs to carry a

large current. A width of the power connection bus **02** is much larger than that of the first leads **03**.
(32) In the present disclosure, the power connection bus **400** can be electrically connected to the driving chip via the plurality of first fan-out leads **500** in the display panel **000**. Therefore, the current carried by the power connection bus **400** is equal to that carried by the first fan-out lead **500**, and a width of the power connection bus **400** is approximately equal to that of the first fan-out lead **500**. A ratio of the width of the power connection bus **400** to a maximum width of the first fan-out lead **500** ranges from 0.7 to 1.3. In other words, the width of the power connection bus **400** in the present disclosure is less than that of the power connection bus **02** in the related art.

(33) In the present disclosure, a number of the second fan-out leads **600** between any two adjacent first fan-out leads **500** are the same in the fan-out region in the non-display region **000b** of the display panel **000**.

(34) In the embodiment of the present disclosure, referring to FIG. 4 and FIG. 6, FIG. 6 is a schematic structural diagram of films of the display panel shown in FIG. 4. In the display panel **000**, the first power signal line **200** and the power connection bus **400** are disposed at different layers. The first fan-out lead **500** in the display panel **000** includes a first sub-lead **501** and a second sub-lead **502** electrically connected to the first sub-lead **501**. The first sub-lead **501** and the first power signal line **200** are disposed at a same layer, and the second sub-lead **502** and the power connection bus **400** are disposed at a same layer. In this way, a cross-sectional area of the first fan-out lead **500** can be enlarged by disposing the first sub-lead **501** and the second sub-lead **502** connected to the first sub-lead **501** at different layers, such that the first fan-out lead **500** has a higher current carrying capability.

(35) For example, the first sub-lead **501** is made of the same material as the first power signal line **200**, such that the first sub-lead **501** and the first power signal line **200** can be formed by one same patterning process; and the second sub-lead **502** is made of the same material as the power connection bus **400**, such that the second sub-lead **502** and the power connection bus **400** can also be formed by one same patterning process. As shown in FIG. 6, the display panel **000** further includes a first conductive layer **101**, an insulation layer **102**, and a second conductive layer **103**. The first conductive layer **101**, insulation layer **102** and second conductive layer **103** are disposed on a substrate **100** and are stacked in a direction perpendicular to and distal from the substrate **100**. The second sub-lead **502** is included in the first conductive layer **101**. The first sub-lead **501** is included in the second conductive layer **103**. The insulation layer **102** is provided with a via hole a, such that the first sub-lead **501** in the second conductive layer **103** can be electrically connected to a second sub-lead **502** in the first conductive layer **101** via the via hole a.

(36) Optionally, as shown in FIG. 6, an orthographic projection of the first sub-lead **501** on the substrate **100** falls within an orthographic projection of the second sub-lead **502** on the substrate **100**. In this way, the width of the first fan-out lead **500** is reduced, and the width of the non-display region **100b** in the display panel **000** is further reduced, thereby increasing the screen-to-body ratio of the display device prepared by using the display panel **000**.

(37) In the embodiment of the present disclosure, any two adjacent second fan-out leads **600** in the plurality of second fan-out leads **600** are disposed at different layers, and orthographic projections, on the substrate **100**, of any two adjacent second fan-out leads **600** disposed at different layers are staggered. In this way, a distance between the second fan-out leads **600** disposed in the non-display region **100b** can be reduced, the width of the non-display region **100b** in the display panel **000** is further reduced. A manufacturing difficulty of the second fan-out leads **600** is lowered as the distance between the second fan-out leads **600** is increased. In addition, because orthographic projections, on the substrate **100**, of two adjacent second fan-out leads **600** disposed at different layers are staggered, the overlap between the two adjacent second fan-out leads **600** is avoided. Thus, a capacitance value of a parasitic capacitance generated between the two adjacent second fan-out leads **600** is reduced, thereby reducing the degree of influence of the parasitic capacitance on the display effect of the display panel **000**.

(38) Optionally, as shown in FIG. 6, the second fan-out leads **600** in the display panel **000** includes a plurality of third sub-leads **601** and a plurality of fourth sub-leads **602**. The plurality of third sub-leads **601** and the plurality of drive signal lines **300** are disposed at a same layer and the plurality of fourth sub-leads **602** and the plurality of drive signal lines **300** are disposed at different layers, wherein the plurality of third sub-leads **601** and the plurality of fourth sub-leads **602** are arranged in a staggered fashion. In this way, any two adjacent second fan-out leads **600** disposed in the non-display region **100b** may be disposed at different layers. For example, the third sub-lead **601** is made of the same material as the first sub-lead **501** and the first power signal line **200**, such that the third sub-lead **601**, the first sub-lead **501** and the first power signal line **200** can be formed by one same patterning process; and the fourth sub-lead **602** is made of the same material as the second sub-lead **502** and the power connection bus **400**, such that the fourth sub-lead **602**, the second sub-lead **502** and the power connection bus **400** can be formed by one same patterning process. For example, the fourth sub-lead **602** is included in the first conductive layer **101**; and the third sub-lead **601** is included in the second conductive layer **103**.

(39) In the present disclosure, referring to FIG. 7, which is a partially enlarged view of the display panel shown in FIG. 4 at C. The display panel **000** further include a plurality of columns of pixels **800** disposed in the display region **100a**. Each pixel **800** is composed of at least two sub-pixels **801** arranged continuously, and thus each column of pixels **800** may include at least two columns of sub-pixels **801**. The plurality of columns of pixels **800** are in a one-to-one correspondence with the plurality of first power signal lines **200**, and the first power signal line **200** is electrically connected to a plurality of sub-pixels **801** in the column of pixels **800** corresponding to the first power signal line **200**. For example, the plurality of sub-pixels **801** in each pixel **800** includes at least two of a red sub-pixel, a blue sub-pixel, a green sub-pixel, or a white sub-pixel. The embodiment of the present disclosure is schematically described by taking a fact that each pixel **800** includes a red sub-pixel, a blue sub-pixel, a green sub-pixel, and a white sub-pixel as an example.

(40) The plurality of drive signal lines **300** in the display panel **000** may include a plurality of data signal lines **301** and a plurality of sense signal lines **302**, wherein the plurality of data signal lines **301** is in a one-to-one correspondence with the plurality of columns of sub-pixels **801** in the display panel **000**, and the plurality of sense signal lines **302** is in a one-to-one correspondence with the plurality of columns of pixels **800**. The data signal line **301** is electrically connected to a plurality of sub-pixels in the column of sub-pixels **801** corresponding to the data signal line **301**. The sense signal line **302** is electrically connected to a plurality of sub-pixels in the column of pixels **800** corresponding to the sense signal line **302**.

(41) For example, as shown in FIG. 4 and FIG. 7, the plurality of data signal lines **301** includes a plurality of groups of data signal lines **301**, one group of data signal lines **301** include two adjacent data signal lines **301**, and one first power signal line **200** or one sense signal line **302** may be arranged between any two adjacent groups of data signal lines **301**.

(42) In the embodiment of the present disclosure, referring to FIG. 4, the display panel **000** further includes a second power signal line **1100** disposed in the non-display region **100b**; and the second power signal line **1100** is disposed on one side, proximal to the power connection bus **400**, of the display region **100a**. The second power signal line **1100** is configured to be electrically connected to the driving chip. In this way, the second power signal line **1100** may be connected to a same driving chip with the first fan-out lead **500** and the second fan-out lead **600**, such that there is no need to provide another driving chip in another region of the display panel **000**, and thus a width of the other non-display region around the display panel **000** can be reduced, thereby reducing a width of an entire frame of the display panel **000**.

(43) For example, the second power signal line **1100** and the power connection bus **400** are disposed at a same layer. For example, the power connection bus **400** is made of the same material as the second power signal line **1100**, such that the second power signal line **1100** and the power connection bus **400** can also be formed by one same patterning process.

(44) In the embodiment of the present disclosure, each sub-pixel **801** in the display panel **000** includes a pixel driving circuit and a light-emitting device. For example, the light-emitting device may be an OLED light-emitting device. The OLED light-emitting device may include an anode, a light-emitting layer, and a cathode that are laminated. As shown in FIG. 8, FIG. 8 is a schematic diagram of a pixel drive circuit according to an embodiment of the present disclosure. The pixel drive circuit includes a first thin-film transistor **T1**, a second thin-film transistor **T2**, a third thin-film transistor **T3**, and a storage capacitor **Cst**.

(45) The pixel drive circuit has an input terminal **D**, a first gate scanning input terminal **G1**, a second gate scanning input terminal **G2**, a first power terminal **VDD**, a second power terminal **VSS**, and a sense signal terminal **Sense**. The input terminal **D** is electrically connected to a data signal line **301**; the first gate scanning input terminal **G1** and the second gate scanning input terminal **G2** are electrically connected to two gate lines in the display panel, respectively; the first power terminal **VDD** is electrically connected to the first power signal line **200**; the second power terminal **VSS** is electrically connected to the second power signal line **1100**; and the sense signal terminal **Sense** is electrically connected to the sense signal line **302**.

(46) Each thin-film transistor may include a first electrode, a second electrode, and a gate. The first electrode is one of a source and a drain, and the second electrode is the other of the source and the drain. A first electrode of the first thin-film transistor **T1** is electrically connected to the first power terminal **VDD**; a second electrode of the first thin-film transistor **T1** is electrically connected to the anode of an OLED light-emitting device **L**; and a gate of the first thin-film transistor **T1** is electrically connected to a first electrode of the second thin-film transistor **T2**. A second electrode of the second thin-film transistor **T2** is electrically connected to the input terminal **D**; and a gate of the second thin-film transistor **T2** is electrically connected to the first gate scanning input terminal **G1**. A first electrode of the third thin-film transistor **T3** is electrically connected to the anode of the OLED light-emitting device **L**; a second electrode of the third thin-film transistor **T3** is electrically connected to the sense signal terminal **Sense**; and a gate of the third thin-film transistor **T3** is electrically connected to the second gate scanning input terminal **G2**. One end of the storage capacitor **Cst** is electrically connected to the first electrode of the second thin-film transistor **T2**, and the other end is electrically connected to the anode of the OLED light-emitting device **L**. The cathode of the OLED light-emitting device **L** is electrically connected to the second power terminal **VSS**.

(47) Optionally, a thin-film transistor in each pixel drive circuit may be a top-gate thin-film transistor, a bottom-gate thin-film transistor, a back channel-etched (BCE) thin-film transistor or an etch stop layer (ESL) thin-film transistor.

(48) In the embodiment of the present disclosure, the thin-film transistor in the pixel drive circuit needs to be shielded by a conductive light-shielding layer, so as to avoid a shift of a voltage threshold value of an active layer in the thin-film transistor under irradiation of light.

(49) Optionally, a material of the conductive light-shielding layer may include a metal material such as metal aluminum, metal silver, metal molybdenum, or alloy.

(50) In the present disclosure, as shown in FIG. 7, the display panel **000** further includes a first connection line **900** and a second connection line **1000** that are disposed in the display region **100a**. The first connection line **900** and the first power signal line **200** are disposed at different layers, and the second connection line **1000** and the sense signal line **302** are disposed at different layers. The first power signal line **200** is electrically connected to a plurality of sub-pixels **801** in a corresponding column of pixels **800** via the first connection line **900**, and the sense signal line **302** is electrically connected to a plurality of sub-pixels **801** in a corresponding column of pixels **800** via the second connection line **1000**.

(51) For example, the first connection line **900** and the power connection bus **400** are disposed at a same layer, and the second connection line **1000** and the conductive light-shielding layer are disposed at a same layer. For example, the first connection line **900** is made of the same material as

the power connection bus **400**, such that the first connection line **900** and the power connection bus **400** can also be formed by one same patterning process; and the second connection line **1000** is made of the same material as the conductive light-shielding layer, such that the second connection line **1000** and the conductive light-shielding layer can be formed by one same patterning process.

(52) In the embodiment of the present disclosure, referring to FIG. **4**, the display panel **000** further includes an antistatic structure **700** disposed in the non-display region **100b**, and the antistatic structure **700** is disposed on one side, distal from the power connection bus **400**, of the display region **100a**. In this way, the antistatic structure **700** and the power connection bus **400** are respectively disposed on two sides of the display region **100a** in the display panel **000**, and a width of a side, in which the power connection bus **400** is disposed, of the non-display region **100b** is further reduced, thereby reducing a width of a side, in which the power connection bus **400** is disposed, of a frame of the display panel **000**.

(53) For example, referring to FIG. **9**, FIG. **9** is a schematic structural diagram of an antistatic structure according to an embodiment of the present disclosure. The antistatic structure **700** may be an electro-static discharge (ESD) diode. The antistatic structure **700** may include a plurality of ESD diodes in a one-to-one correspondence with a plurality of columns of sub-pixels **801**, and each of the ESD diodes **700** may include a first electrode **701**, a second electrode **702**, an active layer **703**, and a third electrode **704**. The first electrode **701** and the second electrode **702** are disposed at a same layer, the first electrode **701** and the active layer **703** are disposed at different layers, and the active layer **703** and the third electrode **704** are disposed at different layers. The first electrode **701** is electrically connected to a data signal line **301** corresponding to a corresponding column of sub-pixels **801**, the second electrode **702** is electrically connected to a second power signal line **1100**, and both the first electrode **701** and the second electrode **702** are electrically connected to the active layer **703**.

(54) When static electricity is generated during use of the display panel **000**, an electrostatic charge is transferred to the first electrode **701** in the ESD diode **700** via the data signal line **301**, such that a coupling capacitance is generated between the first electrode **701** and the third electrode **704**. When accumulated charges on the third electrode **701** reach an on-current value of the ESD diode, the third electrode **704** may turn on the active layer **703** to discharge the accumulated electrostatic charges to the second power signal line **1100** via the second electrode **702**, such that the electrostatic charges generated in the display panel **000** are discharged and the display panel **000** is protected.

(55) For example, the first electrode **701** and the second electrode **702** are disposed at a same layer as a drive signal line **300**, and the third electrode **704** and the second power signal line **1100** are disposed at a same layer. For example, the first electrode **701** and the second electrode **702** are made of the same material as the drive signal line **300**, such that the first electrode **701**, the second electrode **702** and the drive signal line **300** can be formed by one same patterning process; and the third electrode **704** is made of the same material as the second power signal line **1100**, such that the third electrode **704** and the second power signal line **1100** can be formed by one same patterning process.

(56) In an embodiment of the present disclosure, please refer to FIG. **10**, which is a partially enlarged view of the display panel shown in FIG. **5** at D. The drive signal line **300** and the power connection bus **400** are disposed at different layers. The drive signal line **300** includes a drive signal line body **300a** and a jumper signal line **300b** connected to the drive signal line body **300a**. An overlapping area exists between an orthographic projection of the jumper signal line **300b** on the substrate **100** and an orthographic projection of the power connection bus **400** on the substrate **100**, and a width of the jumper signal line **300b** is larger than a width of the drive signal line body **300a**. Thus, when the width of the jumper signal line **300b** in the drive signal line **300** for overlapping with the power connection bus **400** is larger, a risk of disconnection at the overlap of the drive signal line **300** and the power connection bus **400** can be effectively reduced.

(57) Optionally, referring to FIG. 5, an extension direction of the first power signal line **200** in the display panel **000** is parallel to an extension direction of the drive signal line **300** and is perpendicular to an extension direction of the power connection bus **400**, which facilitates the design and manufacture of the display panel **000**, and avoids a short circuit between the first power signal line **200** and the drive signal line **300**. In addition, the first power signal line **200** can be electrically connected to the power connection bus **400** easily.

(58) It should be noted that one patterning process in the above embodiments may include photoresist coating, exposure, development, etching, and photoresist stripping.

(59) In summary, the display panel according to the embodiment of the present disclosure includes a substrate, a plurality of first power signal lines, a plurality of drive signal lines, a power connection bus, a plurality of first fan-out leads, and a plurality of second fan-out leads. Any one of the first fan-out leads in the display panel is disposed between two adjacent second fan-out leads, and a region in which the plurality of second fan-out leads are disposed may also be referred to as a fan-out region. Therefore, the first fan-out leads are also disposed in the fan-out region, and the power connection bus in the display panel is electrically connected to the driving chip via the plurality of first fan-out leads disposed in the fan-out region. In this way, the current carried by the power connection bus is reduced, such that a width of the power connection bus is reduced, a width of the non-display region in the display panel is reduced, and further a width of the frame of the display panel is reduced, thereby increasing a screen-to-body ratio of the display device prepared by using the display panel and improving the display effect of the display device.

(60) An embodiment of the present disclosure further provides a method for manufacturing a display panel. The method for manufacturing the display panel is used for manufacturing the display panel shown in FIG. 2. The method for manufacturing the display panel may include the following steps.

(61) In step A1, a substrate is provided, wherein the substrate is provided with a display region and a non-display region disposed at a periphery of the display region.

(62) In step A2, a plurality of first power signal lines and a plurality of drive signal lines are formed in the display region, and a power connection bus, a plurality of first fan-out leads and a plurality of second fan-out leads are formed in the non-display region.

(63) The power connection bus is electrically connected to the first power signal line and the first fan-out lead respectively; the plurality of second fan-out leads are electrically connected to the plurality of drive signal lines in a one-to-one correspondence; any one of the first fan-out leads is disposed between two adjacent second fan-out leads; and the plurality of first fan-out leads and the plurality of second fan-out leads are both configured to be electrically connected to a driving chip.

(64) A person skilled in the art may clearly understand that, for convenience and conciseness of description, for the specific principle of the display panel described above, reference may be made to the corresponding content in the foregoing embodiment of the structure of the display device. Details are not described herein again.

(65) In summary, according to the method for manufacturing a display panel in the embodiment of the present disclosure, any one of the first fan-out leads in the display panel is disposed between two adjacent second fan-out leads, and a region in which the plurality of second fan-out leads are disposed may also be referred to as a fan-out region. Therefore, the first fan-out leads are also disposed in the fan-out region, and the power connection bus in the display panel is electrically connected to the driving chip via the plurality of first fan-out leads disposed in the fan-out region. In this way, the current carried by the power connection bus is reduced, such that a width of the power connection bus is reduced, a width of the non-display region in the display panel is reduced, and further a width of the frame of the display panel is reduced, thereby increasing a screen-to-body ratio of the display device prepared by using the display panel and improving the display effect of the display device.

(66) An embodiment of the present disclosure further provides a display device. The display device

may be any product or component having a display function such as a mobile phone, a tablet computer, a television set, a display, a notebook computer, a digital photo frame, a navigator, or the like. For example, as shown in FIG. 11, FIG. 11 is a schematic structural diagram of a display device according to an embodiment of the present disclosure. The display device **010** includes a driving chip **001** and a display panel **000**. The display panel **000** may be the display panel in the embodiments described above. The driving chip **001** is electrically connected to the plurality of first fan-out leads and the plurality of second fan-out leads in the display panel **000**. The driving chip **001** may also be electrically connected to the second power signal line. In the display device **010** provided in the embodiment of the present disclosure, there are two driving chips **001**. In another possible embodiment, there may be one or more driving chips **001**, which is not limited in the embodiment of the present disclosure.

(67) When the display device is a television, referring to FIG. 12 and FIG. 13. FIG. 12 is an effect diagram of a display device **020** according to the related art; and FIG. 13 is an effect diagram of a display device according to an embodiment of the present disclosure. Compared with the related art, a width of the frame of the display device **010** provided in the embodiment of the present disclosure is smaller. In the related art, a width of the frame of the display device **020** is 10 mm. In the embodiment of the present disclosure, the width of the frame of the display device **010** is 5 mm. Compared with the related art, the width of the frame of the display device **010** provided in the embodiment of the present disclosure is reduced by 50%, thereby effectively improving the display effect of the display device **010**.

(68) An embodiment of the present disclosure further provides a splicing screen. The splicing screen may include a plurality of display devices **010** shown in FIG. 11. For example, referring to FIG. 14 and FIG. 15, FIG. 14 is an effect diagram of a splicing screen according to the related art, and FIG. 15 is an effect diagram of a splicing screen according to an embodiment of the present disclosure. The splicing screen according to the embodiment of the present disclosure is composed of a plurality of display devices **010** shown in FIG. 10, and a width of the frame of the display device **010** is small. Therefore, compared with the related art, a width of a splicing seam between any two adjacent display devices of the splicing screen according to the embodiment of the present disclosure is smaller, thereby improving the display effect of the splicing screen.

(69) It should be noted that in the drawings, dimensions of a layer and a region may be exaggerated for clarity of illustration. It may also be understood that when an element or a layer is above another element or layer, it may be directly on another element, or there may be an intermediate layer. In addition, it may be understood that when an element or a layer is below another element or layer, it may be directly under another element or there may be more than one intermediate layer or element. In addition, it may be also understood that when a layer or an element is between two layers or two elements, the layer may be a unique layer between two layers or between two elements, or there may be more than one intermediate layer or element. Similar reference numerals throughout the present disclosure indicate similar elements.

(70) In the present disclosure, the terms “first” and “second” are merely intended for a purpose of description, and shall not be understood as an indication or implication of relative importance. The term “a plurality of” means two or more than two unless otherwise specified.

(71) The foregoing descriptions are merely optional embodiments of the present disclosure, but are not intended to limit the present disclosure. Any modification, equivalent replacement, improvement, or the like made without departing from the spirit and principle of the present disclosure shall fall within the protection scope of the present disclosure.

Claims

1. A display panel, comprising: a substrate provided with a display region and a non-display region disposed at a periphery of the display region; a plurality of first power signal lines and a plurality of

drive signal lines that are disposed in the display region; and a power connection bus, a plurality of first fan-out leads, and a plurality of second fan-out leads that are disposed in the non-display region, wherein the power connection bus is electrically connected to the plurality of first power signal lines and the plurality of first fan-out leads respectively; the plurality of second fan-out leads are electrically connected to the plurality of drive signal lines in a one-to-one correspondence; each first fan-out lead is disposed between every two adjacent second fan-out leads; and the plurality of first fan-out leads and the plurality of second fan-out leads are both configured to be electrically connected to a driving chip, wherein an extension direction of each first power signal line is parallel to an extension direction of each drive signal line, and is perpendicular to an extension direction of the power connection bus.

2. The display panel according to claim 1, wherein n first power signal lines correspond to one first fan-out lead of the plurality of first fan-out leads, and the one first fan-out lead is configured to apply a first power signal to the n first power signal lines corresponding to the one first fan-out lead via the power connection bus, wherein n is an integer greater than or equal to 1.

3. The display panel according to claim 2, wherein on the power connection bus, a number of second connection points distributed on a side of a first connection point is equal to a number of second connection points distributed on another side of the first connection point, wherein the first connection point is a connection point for connecting the one first fan-out lead of the plurality of first fan-out leads to the power connection bus; and the second connection point is a connection point for connecting the n first power signal lines corresponding to the one first fan-out lead of the plurality of first fan-out leads to the power connection bus.

4. The display panel according to claim 3, wherein n is equal to 4.

5. The display panel according to claim 1, wherein a ratio of a width of the power connection bus to a width of the widest fan-out lead of the plurality of first fan-out leads ranges from 0.7 to 1.3.

6. The display panel according to claim 1, wherein each first fan-out lead comprises a first sub-lead and a second sub-lead electrically connected to the first sub-lead; wherein the first sub-lead and the plurality of first power signal lines are disposed at a same layer, and the second sub-lead and the power connection bus are disposed at a same layer, the plurality of first power signal lines and the power connection bus being disposed at different layers.

7. The display panel according to claim 6, wherein an orthographic projection of the first sub-lead on the substrate falls within an orthographic projection of the second sub-lead on the substrate.

8. The display panel according to claim 1, wherein two adjacent second fan-out leads are disposed at different layers; and orthographic projections, on the substrate, of the two adjacent second fan-out leads disposed at different layers are staggered.

9. The display panel according to claim 8, wherein the plurality of second fan-out leads comprise a plurality of third sub-leads and a plurality of fourth sub-leads, wherein the plurality of third sub-leads and the plurality of drive signal lines are disposed at a same layer, the plurality of fourth sub-leads and the plurality of drive signal lines are disposed at different layers, the plurality of third sub-leads and the plurality of fourth sub-leads being arranged in a staggered fashion.

10. The display panel according to claim 1, wherein the display panel further comprises an antistatic structure disposed in the non-display region, the antistatic structure being disposed on one side, distal from the power connection bus, of the display region.

11. The display panel according to claim 1, wherein the display panel further comprises a plurality of columns of pixels disposed in the display region, each column of pixels comprise at least two columns of sub-pixels, the plurality of columns of pixels are in a one-to-one correspondence with the plurality of first power signal lines, and each first power signal line is electrically connected to a plurality of sub-pixels in a corresponding column of pixels; and the plurality of drive signal lines comprise: a plurality of data signal lines in a one-to-one correspondence with a plurality of columns of sub-pixels in the display panel and a plurality of sense signal lines in a one-to-one correspondence with the plurality of columns of pixels, wherein each data signal line is electrically

- connected to a plurality of sub-pixels in a corresponding column of sub-pixels, and each sense signal line is electrically connected to a plurality of sub-pixels in a corresponding column of pixels.
12. The display panel according to claim 11, wherein the display panel further comprises a first connection line and a second connection line that are disposed in the display region; wherein the first connection line and the plurality of first power signal lines are disposed at different layers; and the second connection line and the plurality of sense signal lines are disposed at different layers; each first power signal line being electrically connected to the plurality of sub-pixels in the corresponding column of pixels via the first connection line, and each sense signal line being electrically connected to the plurality of sub-pixels in the corresponding column of pixels via the second connection line.
13. The display panel according to claim 1, wherein the display panel further comprises a second power signal line disposed in the non-display region, wherein the second power signal line is disposed on one side, proximal to the power connection bus, of the display region, the second power signal line being configured to be electrically connected to the driving chip.
14. The display panel according to claim 1, wherein the plurality of data signal lines comprise a plurality of groups of data signal lines, wherein each group of data signal lines comprise two adjacent data signal lines, and one first power signal line or one sense signal line is disposed between any two adjacent groups of data signal lines.
15. The display panel according to claim 1, wherein the plurality of drive signal lines and the power connection bus are disposed at different layers, wherein each drive signal line comprises a drive signal line body and a jumper signal line connected to the drive signal line body, an overlapping area exists between an orthographic projection of the jumper signal line on the substrate and an orthographic projection of the power connection bus on the substrate, and a width of the jumper signal line is larger than a width of the drive signal line body.
16. A method for manufacturing a display panel, comprising: providing a substrate, wherein the substrate is provided with a display region and a non-display region disposed at a periphery of the display region; and forming a plurality of first power signal lines and a plurality of drive signal lines in the display region, and forming a power connection bus, a plurality of first fan-out leads and a plurality of second fan-out leads in the non-display region, wherein the power connection bus is electrically connected to the plurality of first power signal lines and the plurality of first fan-out leads respectively; the plurality of second fan-out leads are electrically connected to the plurality of drive signal lines in a one-to-one correspondence; each first fan-out lead is disposed between every two adjacent second fan-out leads; and the plurality of first fan-out leads and the plurality of second fan-out leads are both configured to be electrically connected to a driving chip, wherein an extension direction of each first power signal line is parallel to an extension direction of each drive signal line, and is perpendicular to an extension direction of the power connection bus.
17. A display device, comprising: a driving chip and a display panel, wherein the display panel comprises: a substrate provided with a display region and a non-display region disposed at a periphery of the display region; a plurality of first power signal lines and a plurality of drive signal lines that are disposed in the display region; and a power connection bus, a plurality of first fan-out leads, and a plurality of second fan-out leads that are disposed in the non-display region, wherein the power connection bus is electrically connected to the plurality of first power signal lines and the plurality of first fan-out leads respectively; the plurality of second fan-out leads are electrically connected to the plurality of drive signal lines in a one-to-one correspondence; each first fan-out lead is disposed between every two adjacent second fan-out leads; and the driving chip is electrically connected to the plurality of first fan-out leads and the plurality of second fan-out leads in the display panel, wherein an extension direction of each first power signal line is parallel to an extension direction of each drive signal line, and is perpendicular to an extension direction of the power connection bus.
18. The display device according to claim 17, wherein n first power signal lines correspond to one

first fan-out lead of the plurality of first fan-out leads, and the one first fan-out lead is configured to apply a first power signal to the n first power signal lines corresponding to the one first fan-out lead via the power connection bus, wherein n is an integer greater than or equal to 1.

19. The display device according to claim 17, wherein a ratio of a width of the power connection bus to a width of the widest fan-out lead the plurality of first fan-out leads ranges from 0.7 to 1.3.
